1. Title

V-Shuttle
Sensor Fusion and Human-Centered Dashboard for Autonomous Shuttle Decision Support

2. Project Overview

V-Shuttle is a deterministic sensor-fusion and semantic-parsing system designed to support the safety driver of an autonomous shuttle.

Autonomous vehicles operating in complex environments such as historic city centers may encounter degraded or ambiguous road signs. This can lead to unnecessary emergency braking (phantom braking) and confusion for the safety driver.

Our system merges multiple sensor readings, corrects OCR errors, interprets the meaning of road signs, and presents a clear decision through an intuitive dashboard designed for fast human understanding.

3. Problem

Waymo's autonomous shuttle operates in complex environments with narrow streets and restricted traffic zones (ZTL).

When road signs are degraded, partially occluded, or interpreted inconsistently by sensors, the vehicle may fail to confidently interpret them.

This often results in:

- Phantom braking
- Delays in manual intervention
- Confusing technical logs displayed to the safety driver

The challenge is to bridge the gap between machine perception and human understanding.

4. Solution

V-Shuttle introduces two key components:

1. A deterministic **sensor fusion and semantic parsing engine**
2. A **human-centered live dashboard**

The backend system merges sensor readings using weighted reliability, cleans OCR errors, and interprets the meaning of the road sign.

The frontend dashboard then presents the resulting decision in a simple and intuitive way, allowing the safety driver to quickly understand the situation and act if necessary.

5. Sensor Fusion Logic

Each sensor is assigned a reliability coefficient:

- Front Camera → 1.0
- Side Camera → 0.85
- V2I Receiver → 0.75

The coefficient is multiplied by the confidence score provided by each sensor.

The system then:

1. Normalizes OCR-corrupted text
2. Corrects common recognition errors
3. Computes similarity between sensor readings
4. Selects the most probable interpretation
5. Calculates a final confidence score

This approach avoids a simple average and implements a weighted sensor fusion strategy based on the reliability of the sources.

6. Decision Logic

Based on the fused interpretation and final confidence score, the system determines the action to take.

Possible actions:

- **GO** → the vehicle can proceed
- **STOP** → the vehicle must stop
- **ATTENTION / LOW CONFIDENCE** → the system signals uncertainty

If uncertainty remains and no operator intervention occurs, the system falls back to **STOP** to ensure safety.

7. Dashboard

The dashboard is designed for a safety driver operating under time pressure.

Key features:

- START SIMULATION button
- Automatic scenario progression every 4 seconds
- Color-coded interface (green, yellow, red)
- Large interaction buttons for easy use on a tablet
- Clear visualization of the fused road sign text
- Confidence indicator
- No technical logs or complex debugging messages

8. Architecture

Input JSON Scenarios
↓
Backend Processing
- OCR normalization
- Sensor fusion
- Semantic parsing
- Decision engine
↓
API Response
↓
Frontend Dashboard
- Live simulation
- Color-based feedback
- Human interaction

9. Edge Cases

The system is designed to handle multiple real-world edge cases:

- OCR corrupted text
- Conflicting sensor readings

- Missing or null sensor values
- Offline V2I receiver
- Ambiguous road signs
- Exceptions such as "ECCETTO BUS"
- Time-based restrictions

10. How to Run

Follow these steps to run the project locally.

1. Clone the repository

```
git clone https://github.com/your-repo/v-shuttle.git

cd v-shuttle
```

2. Install dependencies

```
npm install
```

3. Start the application

```
npm run dev
```

4. Open the application

Once the server is running, open your browser and go to:

http://localhost:3000

5. Start the simulation

On the main screen click:

START SIMULATION

The dashboard will automatically start processing scenarios from:

VShuttle-input.json

Every 4 seconds the system loads a new scenario and simulates the interpretation of road signs using sensor fusion and semantic parsing.

11. Screenshots

Dashboard – Normal Operation

Dashboard – Low Confidence Warning